$\therefore e = \dfrac{\text{emissive power of body}}{\text{emissive power of perfect black body.}} = \dfrac{E}{E_b}$ or $\dfrac{P}{A \cdot t} = e$

Emissivity has no unit and dim^n. Value of e depends on the nature of the surface of the emitting body. It ranges from 0 to 1.

For a perfectly black body, $e = 1$

" a black body $e < 1$

For a perfectly reflecting surface $e = 0$.

$\therefore P =$

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Stefan's-Boltzmann's law of black body radiation:

Stefan's declared experimentally that energy radiated by a body depends only on its temp. and he put forward a law known as Stefan's law.

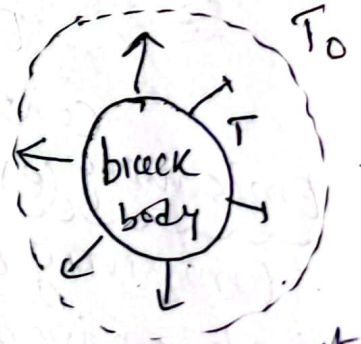
Stefan's-Boltzmann law states that " emissive power of a black body is directly proportional to the fourth power of absolute temp. of surface of the body".

let E be the emissive power of a black body at absolute temp. T. Then, according to Stefan's-Boltzmann law $E \propto T^4$

or $E = \sigma T^4$ — (1) where σ is proportionality constant is called Stefan's Const. The value of σ in SI system is $5.67 \times 10^{-8}\ Wm^{-2}K^{-4}$.

If a body is not perfectly black, and has its emissivity e. Then,

$E = e\sigma T^4$. — (2)



[Fig: Stefan's law of black body radiation

or $\frac{Q}{A \cdot t} = e\sigma T^4$ or $\frac{Q/t}{A} = e\sigma T^4 \Rightarrow \frac{\text{power}}{A} = e\sigma T^4$

$\therefore P = eA\sigma T^4$ — (3)

This relations given above are valid only if the temp. of the surrounding is zero kelvin. But, if there is a surrounding at temp T_0 (more than 0K.) then, surrounding also emits heat radiation, which is absorbed by black body. Hence, relation is modified by Boltzmann for the net emissive power as

$E' = \sigma(T^4 - T_0^4)$ — (4) This is called Stefan's-Boltzmann law.

If a body is not perfect black body, the net emissive power $E' = e\sigma(T^4 - T_0^4)$

$$\frac{Q'}{A \times t} = e\sigma(T^4 - T_0^4)$$

$$\text{or } \frac{Q'/t}{A} = e\sigma(T^4 - T_0^4)$$

$\therefore P' = eA\sigma(T^4 - T_0^4)$ — (5)

This gives power of radiated energy of a black body.

<u>Some Numerical problems</u>

(1) An insulating material having a thermal conductive of 0.08 W/m-K is used to limit the heat transfer of 80 W/m2 for a temp. of 15°C across the opposite faces. Find the thickness of the material.

Sol^n: $K = 0.08$ W/m-K, $\frac{Q}{tA} = 80$ W/m^2 $\Delta T = 15°C = 15K$

$dx = ?$ $\Rightarrow \frac{Q}{t} = KA\frac{dT}{dx} \Rightarrow dx = \frac{Q}{tAKdT} \Rightarrow dx = 0.15m$